Miguel Zumalacárregui

Group Leader, Astrophysical and Cosmological Relativity — Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam
Ramón y Cajal Fellow — Instituto de Física Teórica IFT-UAM/CSIC, Madrid

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The publication record is maintained on INSPIRE-HEP; research projects and funding are listed on the website.

Positions

- **Ramón y Cajal Fellow**
  Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2026–
- **Group Leader**, Astrophysical and Cosmological Relativity
  Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam · 2020–
- **Marie Skłodowska-Curie Global Fellow**
  Berkeley Center for Cosmological Physics, with return phase at IPhT Saclay · 2017–2020
- **Nordita Fellow**
  Nordic Institute for Theoretical Physics, Stockholm · 2015–2017
- **Postdoctoral Researcher**
  Institut für Theoretische Physik, University of Heidelberg · 2013–2015
- **Postdoctoral Researcher**
  Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2013

Education

- **PhD in Physics**, *Cum Laude* with International Mention
  Universidad Autónoma de Madrid · 2012 · Graduate work at ICC Barcelona and IFT-UAM/CSIC (FPI fellowship)
  Dissertation: Probing the Foundations of the Standard Cosmological Model · defense presentation
- **Master's thesis**
  Universidad Autónoma de Madrid · PDF

Supervision

Postdoctoral researchers

- **Han-Gil Choi** — 2026–
- **Xikai Shan** — Affiliated postdoctoral researcher, 2026–
- **Juan Urrutia** — from October 2026
- **Héctor Villarrubia-Rojo** — from October 2026
- **Daniel Ballard** — from October 2026
- **Srashti Goyal** — 2023–2026 → Independent Fellow at IUCAA
- **Serena Giardino** — 2024–2025 → Postdoc at Imperial College London
- **Guilherme Brando** — Humboldt Fellow, 2022–2024 → Tenured Researcher at UFES, Brazil
- **Giovanni Tambalo** — 2020–2023 → Postdoc at ETH Zürich, then Marie Skłodowska-Curie Fellow at IPhT Saclay

PhD students

- **Manon van Zyl** — from October 2026

- **Emanuela Alberici** — from October 2026
- **Nicola Menadeo** — PhD student, U. Naples (co-supervised), 2023–2025, long-term stay → Postdoc at Humboldt University of Berlin
- **Stefano Savastano** — PhD student (co-supervised with A. Buonanno), 2020–2024 → PhD awarded at HU Berlin

Master's students

Pablo Vega del Castillo (LMU, 2026); **Matteo Pagani** (U. Bologna, co-supervised, 2026); **Louis Salimbeni** (ETH Zürich, co-supervised, 2026); **Julius Streibert** (FU Berlin, co-supervised, 2022–2023); **Shashwat Singh** (Paris Observatory, 2022–2023 → PhD at the University of Glasgow); **Richard Stiskalek** (LMU, co-supervised, 2021–2022 → PhD at Oxford, pre-doctoral Fellow at CCA, now postdoc at Oxford); **Alessandro Longo** (U. Trieste, co-supervised, 2020–2021 → PhD student at SISSA).

Research interns and undergraduates

Selin Üstündag (2026 → PhD at Newcastle University); **Marcel Glätzer** (HU Berlin, 2024–2025); **Jaime Raynaud** (TU Berlin & KTH, 2024–2025 → PhD at MPIA); **Vaibhav Omble** (LMU, 2023–2024); **Daryna Yushchenko** (Princeton summer program, 2024); **Lyla Choi** (Princeton summer program, 2023 → PhD at Stanford).

Visiting researchers, and the students mentored informally before the group started in 2020, are listed at miguelzuma.github.io/group.html.

Scientific coordination

- **LISA Consortium**
  member since 2018
  Council member and Senior Co-Chair of the Cosmology Working Group (2025–); section coordinator for gravitational-wave lensing in the LISA Cosmology white paper (2022)
- **Euclid Consortium**
  member since 2014
  Lead of the Science Working Group on Gravitational Waves and of its "bright sirens" work package (2025–); member of the Strong Lensing, Cosmology Theory, and Supernovae & Transients working groups
- **GW-Space 2050**
  Coordinator for the cosmography section, defining ESA's gravitational-wave missions after LISA (2023–)
- **Multi-messenger lensing white paper**
  Coordinator for the microlensing section (2024)

Workshops and meetings organized

- 13th LISA Cosmology Working Group Workshop
  Barcelona, 2026
- 12th LISA Cosmology Working Group Workshop
  Tallinn, 2025
- Lensing and Wave Optics in Strong Gravity
  Erwin Schrödinger Institute, Vienna, 2024
- LISA joint Fundamental Physics + Waveform meeting
  Potsdam, 2024
- ACR Division Seminars
  Max Planck Institute for Gravitational Physics, 2020–2022
- BCCP Cosmology Lunch

UC Berkeley, 2016–2017

- Cosmology Seminars, Institut für Theoretische Physik
  University of Heidelberg, 2014–2015
- TRR33 Winter Schools
  Passo del Tonale, 2013, 2014 and 2015

Teaching

Graduate schools

- **Gravitational waves** — TAE (Taller de Altas Energías), Benasque, 2024
  Slides
- **Tests of gravity and dark energy with cosmology and gravitational waves**
  II Mesoamerican Workshop on Gravity and Cosmology, Chiapas, 2018
- **Alternative gravity theories** — Cosmology School in the Canary Islands, 2017
  Video lecture
- **School for Cosmology Tools** — IFT Madrid, 2017
  Lectures on CLASS, hi_class and MontePython; material on the Talks & Teaching page
- **Workshop on CLASS and MontePython** — MPIA Munich, 2014

University courses

- **Berkeley Connect Instructor** — UC Berkeley, 2018–2019
  Two semesters of undergraduate mentoring and teaching
- **Lecturer** — University of Heidelberg, 2014–2015
  Advanced Cosmology; Introduction to CMB Physics; Statistical Mechanics; advanced seminar
- **Teaching assistant** — University of Heidelberg, 2014–2015
  Theoretical Astrophysics (graduate), with M. Bartelmann
- **Teaching assistant** — Universidad Autónoma de Madrid, 2013
  Gravitation and Cosmology; Physics of the Cosmos
- **Mini-course on inhomogeneous cosmologies** — University of Barcelona, 2012

Awards and fellowships

- ERC Consolidator Grant (2025) — GLOW, Gravitational Lensing of Waves.
- Ramón y Cajal Fellowship (2024 call, awarded 2025).
- Max Planck Society Annual Project (2024) — With Einstein on Crooked Paths.
- R3 certification, Spanish Research Agency.
- Berkeley Connect Fellow, UC Berkeley (2018).
- Marie Skłodowska-Curie Global Fellowship (2015).
- Nordita Fellowship (2015).
- Yggdrasil Fellowship (2011) — research stay at ITA Oslo.
- FPI doctoral fellowship (2008) and UAM Master's scholarship (2007).

Evaluation and peer review

Evaluation panels

- NASA Hubble Fellowship Program, Physics and Cosmology panel.
- FCT — Portuguese Foundation for Science and Technology, Individual Support Call.
- Max Planck Institute for Gravitational Physics — postdoctoral hiring committee, since 2020.

External evaluator for funding agencies

European Research Council (Starting Grants), STFC/UKRI, Swiss National Science Foundation, Alexander von Humboldt Foundation, InterTalentum (Marie Skłodowska-Curie COFUND), Netherlands Research Council (NWO), Research Grants Council of Hong Kong, and the Estonian Research Council.

Refereeing

Peer review for more than 40 papers, including Physical Review Letters, Physical Review D, Nature Astronomy, JCAP, The Astrophysical Journal and Astrophysical Journal Letters.

Selected talks

- **Spanish & Portuguese Relativity Meeting** — Universidad de Murcia, 2026 · plenary
  Gravitational lensing of waves: a new window into astrophysics, dark matter and gravity
- **IAS and Princeton University** — gravitational-wave meeting, Princeton, 2026 · seminar
  GW231123 as a diffracted and magnified black-hole merger
- **Leibniz Institute for Astrophysics** — Potsdam, 2024 · institute colloquium
  Gravitational lensing of waves
- **New Frontiers in Strong Gravity** — Benasque, 2024 · plenary
  Wave-optical phenomena in gravitational-wave lensing
- **Workshop on Tensions in Cosmology** — Corfu, 2022 · plenary
  Towards solutions to the Hubble problem beyond Einstein's gravity
- **EuCAPT Annual Symposium** — CERN, 2022 · plenary review
  Dark energy and tests of gravity with cosmology and gravitational-wave propagation
- **Institute of Theoretical Astrophysics** — University of Oslo, 2021 · institute colloquium
  The dark Universe in the era of gravitational-wave astronomy
- **Progress in Old and New Themes in Cosmology** — Palais des Papes, Avignon, 2020 · plenary
  Tests of gravity with gravitational waves
- **COSMO** — Aachen University, 2019 · plenary
  Tests of dark energy and modified gravity
- **Rencontres du Vietnam** — Very High Energy Phenomena in the Universe, 2018 · plenary
  The dark Universe in the era of gravitational-wave astronomy

Languages

Spanish (native), English (C2), German (C1).

Selected publications

Chosen for what they opened up rather than for where they appeared. The complete record is on INSPIRE-HEP.

- Across the Universe: GW231123 as a magnified and diffracted black hole merger
  S. Goyal, H. Villarrubia-Rojo, M. Zumalacárregui · Astrophys. J. Lett. 1008, L12 (2026)
- Black holes as telescopes: discovering supermassive binaries through quasi-periodic lensed starlight
  H. Wang, M. Zumalacárregui, B. Kocsis · Phys. Rev. Lett. 136, 061403 (2026) · Editors' Suggestion
- GLoW: novel methods for wave-optics phenomena in gravitational lensing
  H. Villarrubia-Rojo, S. Savastano, M. Zumalacárregui, L. Choi, S. Goyal, L. Dai, G. Tambalo · Phys. Rev. D 111, 103539 (2025)
- Gravitational wave lensing as a probe of halo properties and dark matter

G. Tambalo, M. Zumalacárregui, L. Dai, M. Cheung · Phys. Rev. D 108, 103529 (2023)

- Gravitational wave lensing beyond general relativity: birefringence, echoes and shadows
  J. M. Ezquiaga, M. Zumalacárregui · Phys. Rev. D 102, 124048 (2020) · Editors' Suggestion
- Gravity in the era of equality: towards solutions to the Hubble problem without fine-tuned initial conditions
  M. Zumalacárregui · Phys. Rev. D 102, 023523 (2020)
- Limits on stellar-mass compact objects as dark matter from gravitational lensing of type Ia supernovae
  M. Zumalacárregui, U. Seljak · Phys. Rev. Lett. 121, 141101 (2018) · Editors' Suggestion
- Dark energy after GW170817: dead ends and the road ahead
  J. M. Ezquiaga, M. Zumalacárregui · Phys. Rev. Lett. 119, 251304 (2017) · Editors' Suggestion
- hi_class: Horndeski in the Cosmic Linear Anisotropy Solving System
  M. Zumalacárregui, E. Bellini, I. Sawicki, J. Lesgourgues, P. G. Ferreira · JCAP 08, 019 (2017)
- Transforming gravity: from derivative couplings to matter to second-order scalar–tensor theories beyond the Horndeski Lagrangian
  M. Zumalacárregui, J. García-Bellido · Phys. Rev. D 89, 064046 (2014)